Dogyoon Song

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Curriculum vitae
October 25, 2026

Academic Appointments

University of California, Davis Jul 2024–present
Assistant Professor of Statistics and Data Science

University of Michigan, Ann Arbor Jul 2021–Jun 2024
Postdoctoral Research Fellow
Mentors: Alfred O. Hero and Qing Qu

Max Planck Institute for Mathematics in the Sciences Aug–Sep 2021
Visiting Postdoctoral Researcher, Leipzig, Germany
Convexity, Optimization and Data Science Research Group
Mentor: Venkat Chandrasekaran

Education

Massachusetts Institute of Technology, Cambridge, MA Jun 2021
Ph.D. in Electrical Engineering and Computer Science
Advisors: Pablo A. Parrilo and Devavrat Shah
Thesis: *Addressing Missing Data and Scalable Optimization for Data-driven Decision Making*

Massachusetts Institute of Technology, Cambridge, MA Jun 2016
S.M. in Electrical Engineering and Computer Science
Advisor: Devavrat Shah
Thesis: *Blind Regression: Nonparametric Regression for Latent Models via Collaborative Filtering*

Seoul National University, Seoul, South Korea Feb 2013
Major: Electrical and Computer Engineering (B.S.)
Co-majors: Mathematical Sciences (B.S.); Physics (B.S.); Economics (B.A.)

Publications and Manuscripts

(α–β) denotes alphabetical author order; $\star$ denotes equal contribution as co-first authors.

Preprints and Manuscripts in Preparation

M11. **D. Song**, Multiscale dynamics of oscillatory gradient descent in window coordinates.
Manuscript in preparation for conference submission.

M10. **D. Song**, Learning from large-step gradient descent via trajectory post-processing.
Manuscript in preparation for conference submission.

M9. (α–β) J. J. Ryu, **D. Song**, Sharp subspace recovery in self-orthogonalizing streaming principal component analysis.
Manuscript in preparation for journal submission.

M8. L. Yang, P. Ding, D. Shen, **D. Song**, Regression adjustment versus difference-in-means under overparameterization.
Manuscript in preparation for journal submission.

M7. R. Malinas, B. D. Robinson, **D. Song**, A. O. Hero,, High-dimensional quickest change point detection.
Manuscript in preparation for journal submission.

M6. J. Kim, **D. Song**, C. Yun, From geometry to generalization: Why row normalization can beat Adam and Muon.
Under review at a conference. arXiv:2610.11309 (v1, Oct 2026).

M5. S. M. Kwon, A. S. Xu, C. Yaras, **D. Song**, L. Balzano, Q. Qu, The effect of training task diversity on in-context learning through the lens of low-dimensional subspaces.
Under review at a journal. arXiv:2606.06814 (v1, Jun 2026).

M4. **D. Song**, Neumann-series corrections for regression adjustment in randomized experiments.
Under review at a journal. arXiv:2511.08539 (v1, Nov 2025).

M3. **D. Song**, A. O. Hero, On separability of covariance in multiway data analysis.
Under review at a journal. arXiv:2302.02415 (v3, March 2026).

Journal Articles

J6. D. Shen*, **D. Song***, P. Ding, J. Sekhon, Benign overfitting beyond prediction: The ordinary least squares interpolator.
Biometrika, 2026.

J5. (α–β) Y. Chen*, **D. Song***, X. Xi, Y. Zhang, Local minima structures in Gaussian mixture models.
IEEE Transactions on Information Theory, 2024.

J4. **D. Song**, P. A. Parrilo, On approximations of the PSD cone by a polynomial number of smaller-sized PSD cones.
Mathematical Programming, 2023.

J3. Y. Wang, Z. Sun, **D. Song**, A. O. Hero, Kronecker-structured covariance models for multiway data.
Statistics Surveys, 2022.

J2. (α–β) A. Agarwal*, D. Shah, D. Shen*, **D. Song***, On robustness of principal component regression.
Journal of the American Statistical Association, 2021. Earlier conference version: C5.

J1. (α–β) Y. Li, D. Shah, **D. Song***, C. E. Lee Yu* (formerly C. E. Lee), Nearest neighbors for matrix estimation interpreted as blind regression for latent variable model.
IEEE Transactions on Information Theory, 2020. Earlier conference version: C4.

Refereed Conference Papers

C16. **D. Song**, Design-based finite-sample analysis for regression adjustment.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.

C15. J. Kim, **D. Song**, C. Yun, Scaling laws of signSGD in linear regression: When does it outperform SGD?
International Conference on Learning Representations (ICLR), 2026.

C14. R. Malinas, **D. Song**, B. D. Robinson, A. O. Hero, High-dimensional sequential change detection.
IEEE International Symposium on Information Theory, 2025.

C13. R. Malinas, **D. Song**, A. O. Hero, Sequential community detection in high-dimensional temporal graphs.
Asilomar Conference on Signals, Systems, and Computers, 2024.

C12. H. Zhang*, Y. Lu*, I. Alkhouri, S. Ravishankar, **D. Song**, Q. Qu, Improving training efficiency of diffusion models via multi-stage framework and tailored multi-decoder architecture.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

C11. S. Kwon*, Z. Zhang*, **D. Song**, L. Balzano, Q. Qu, Efficient low-dimensional compression of overparameterized models.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2024.

C10. **D. Song***, K. Han*, Errors-in-variables Fréchet regression with low-rank covariate approximation.
Advances in Neural Information Processing Systems (NeurIPS), 2023.

C9. Z. Sun*, **D. Song***, A. O. Hero, Minimum-risk recalibration of classifiers.
Advances in Neural Information Processing Systems (NeurIPS), 2023. **Spotlight**.

C8. R. Malinas, **D. Song**, A. O. Hero, Community detection in high-dimensional graph ensembles.
Asilomar Conference on Signals, Systems, and Computers, 2023.

C7. J. Jia, Y. Zhang, **D. Song**, S. Liu, A. O. Hero, Robustness-preserving lifelong learning via dataset condensation.
IEEE International Conference on Acoustics, Speech and Signal Processing, 2023.

C6. (α–β) D. Shah, **D. Song***, Z. Xu*, Y. Yang, Sample efficient reinforcement learning via low-rank matrix estimation.
Advances in Neural Information Processing Systems (NeurIPS), 2020.

C5. (α–β) A. Agarwal*, D. Shah, D. Shen*, **D. Song***, On robustness of principal component regression.
Advances in Neural Information Processing Systems (NeurIPS), 2019. **Oral presentation**.

C4. (α–β) C. E. Lee*, Y. Li, D. Shah, **D. Song***, Blind regression: Nonparametric regression for latent variable models via collaborative filtering.
Advances in Neural Information Processing Systems (NeurIPS), 2016.

C3. T. Kim, **D. Song**, S. Youn, J. Park, H. Park, J. Kim, Verifying start-up failures in coupled ring oscillators in presence of variability using predictive global optimization.
IEEE/ACM International Conference on Computer-Aided Design, 2013.

C2. J. Kim, J. Lee, **D. Song**, T. Kim, K.-H. Kim, S. Jung, S. Youn, Discretization and discrimination methods for design, verification, and testing of analog/mixed signal circuits.
IEEE Custom Integrated Circuits Conference, 2013.

C1. **D. Song**, J. Kim, A low-power high-radix switch fabric based on low-swing signaling and partially-activated input lines.
International Symposium on VLSI Design, Automation, and Test, 2013.

Earlier Preprints

M2. (α–β) D. Shah, **D. Song**, Deconvolution with unknown error distribution interpreted as blind isotonic regression.
arXiv:1803.03469 (v3, April 2020).

M1. (α–β) D. Shah, **D. Song**, Learning RUMs: Reducing mixture to single component via PCA.
arXiv:1812.11917 (v3, March 2020).

Research Presentations

Regression adjustment in high-dimensions: A design-based finite-sample view — Jun 2026
ACM SIGMETRICS Causal Inference Workshop; Ann Arbor, MI

High-dimensional sequential change detection — Dec 2025
CFE-CMStatistics (Zoom)

Regression adjustment with high-dimensional covariates — Nov 2025
Statistics and Data Science Seminar; University of Illinois Chicago (Zoom)

Regression adjustment with high-dimensional covariates — Aug 2025
International Conference on Econometrics and Statistics (EcoSta); Tokyo, Japan

On geometry of regularized M-estimation and structured model selection — Feb 2025
Mathematics of Data & Decisions Seminar; University of California, Davis

Taming overparameterized models: OLS interpolators, regularization, and beyond Oct 2024
Statistics Department Seminar (STA 290); University of California, Davis

Advancing predictions for informed decisions Feb 2024
Department of Statistics Seminar Series; University of Michigan

Advancing predictions for informed decisions Feb 2024
Department of Industrial and Systems Engineering; Texas A&M University

Advancing predictions for informed decisions Feb 2024
Econometrics and Statistics Colloquium; University of Chicago Booth School of Business

Advancing predictions for informed decisions Jan 2024
Statistics Department Seminar; University of California, Davis

Higher-order geometry of losses, regularizing norms and structured M-estimation Oct 2023
INFORMS Annual Meeting; Phoenix, AZ

Higher-order geometry of losses, regularizing norms and structured M-estimation Oct 2023
SIAM Great Lakes Section Annual Meeting; East Lansing, MI

On Kronecker separability of multiway covariance Feb 2023
Information Theory and Applications (ITA) Workshop; San Diego, CA

Covariance modeling for multiway data Dec 2021
Graduate School of Data Science Seminar; Seoul National University (Zoom)

On approximating positive semidefinite cone using small-sized PSD constraints Aug 2021
Numerical Algebra and Optimization Seminar; Max Planck Institute for Mathematics in the Sciences

Measuring optimality gap in conic programming approximations with Gaussian width Jul 2019
SIAM Conference on Applied Algebraic Geometry; Bern, Switzerland

Learning mixtures with missing entries and its application to rankings Feb 2019
MIT LIDS Student Conference; Cambridge, MA

Monotone matrix estimation ≈ robust deconvolution Oct 2018
Allerton Conference on Communication, Control, and Computing; Monticello, IL

Measuring optimality gap in conic programming approximations with Gaussian width Jul 2018
International Symposium on Mathematical Programming (ISMP); Bordeaux, France

Blind regression: Revisiting collaborative filtering Jan 2016
MIT LIDS Student Conference; Cambridge, MA

Honors and Fellowships

Hewlett Packard Fellowship, MIT School of Engineering	2017–2018
Siebel Scholarship	Class of 2016
Samsung Graduate Fellowship	2014–2019
Advanced Television and Signal Processing Fellowship, MIT EECS	2014–2015
Korea Foundation for Advanced Studies Graduate Fellowship	2013 (declined)
Excellence Award for Outstanding Graduates, SNU School of Engineering	2013

Research Mentoring

University of California, Davis — Assistant Professor

Jihwan Kim, undergraduate, Seoul National University (co-advised with Chulhee Yun). 2025–present
Optimizer scaling laws and normalization geometry (C15, M6).

Letian Yang, Ph.D. student, USC Marshall School of Business (co-advised with Dennis Shen). 2024–present
Regression adjustment in high-dimensional randomized experiments (M7).

University of Michigan — Postdoctoral Research Fellow

Soo Min Kwon, Ph.D. student, Electrical and Computer Engineering.
Low-dimensional compression of overparameterized models (C11, M5).

Huijie Zhang, Ph.D. student, Electrical and Computer Engineering.
Efficient training of diffusion models (C12).

Robert Malinas, Ph.D. student, Electrical and Computer Engineering.
High-dimensional community detection and sequential change detection (C8, C13, C14).

Zeyu Sun, Ph.D. student, Electrical and Computer Engineering.
Classifier recalibration and Kronecker-structured covariance models (C9, J3).

Wayne Wang, Ph.D. student, Statistics.
Kronecker-structured covariance models for multiway data (J3).

Massachusetts Institute of Technology — Doctoral Student

Yihua Li, M.Eng. student, Electrical Engineering and Computer Science.
Matrix estimation and blind regression for latent variable models (C4, J1).

Teaching

University of California, Davis — Instructor

Statistical Data Science III (STA 35C)	Fall 2026; Spring 2026, 2025
Mathematical Statistics: Brief Course (STA 130A)	Fall 2026
Introduction to Probability Theory (STA 131A)	Spring 2026
Theoretical Foundations for Machine Learning (STA 250)	Spring 2025

Massachusetts Institute of Technology — Teaching Assistant

Algebraic Techniques and Semidefinite Optimization (6.256)	Spring 2019
Inference and Information (6.437)	Spring 2018
Nonlinear Programming (6.252)	Spring 2017

Professional and University Service

Conference and Session Organization

Co-organizer, INFORMS Annual Meeting, session on "Geometry and Optimization Methods" Oct 2023

Co-organizer, SIAM Conference on Applied Algebraic Geometry, minisymposium on "Algebraic Methods for Convex Sets" Jul 2019

Reviewing

Journals. Annals of Statistics; Mathematical Programming; IEEE Transactions on Information Theory; SIAM Journal on Applied Algebra and Geometry; Information and Inference.

Conferences. NeurIPS; ICML; ICLR; COLT; AISTATS; Information Theory Workshop (ITW); Conference on Parsimony and Learning (CPAL).

Other. American Mathematical Society (book chapter).

University Service

Ph.D. Admissions Committee, UC Davis Statistics	2024–present
Ph.D. Qualifying Examination Committees, UC Davis Statistics (three students)	2025–2026
Ph.D. Dissertation Committee, UC Davis Statistics (one student; Ph.D. awarded Spring 2025)	2024–2025
Reader for Ph.D. applications, MIT EECS	2017–2018
Career Mentoring Committee, MIT LIDS	2017–2018
Annual Student Conference Organizing Committee, MIT LIDS	2016–2017
Social Committee, MIT LIDS	2015–2016

Other Research Experience

Research Intern, Thomson Reuters Labs	Summer 2017

References

Alfred O. Hero, III
John H. Holland Distinguished University Professor of EECS
R. Jamison and Betty Williams Professor of Engineering
University of Michigan
hero@umich.edu

Pablo A. Parrilo
Joseph F. and Nancy P. Keithley Professor
Department of Electrical Engineering and Computer Science
Massachusetts Institute of Technology
parrilo@mit.edu

Devavrat Shah
Andrew (1956) and Erna Viterbi Professor
Department of Electrical Engineering and Computer Science
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Department of Statistics
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